Mitosis

During the mitotic phase of the cell cycle, an elaborate apparatus ensures that each of the two daughter cells receives a complete set of genetic information. This result is achieved by a mechanism that distributes one chromatid of each chromosome to each daughter cell (Fig. 2-9). The process of distributing a copy of each chromosome to each daughter cell is called **chromosome segregation**. The importance of this process for normal cell growth is illustrated by the observation that many tumors are invariably characterized by a state of genetic imbalance resulting from mitotic errors in the distribution of chromosomes to daughter cells.

The process of mitosis is continuous, but five stages, illustrated in Figure 2-9, are distinguished: prophase, prometaphase, metaphase, anaphase, and telophase.

- ***Prophase.*** This stage is marked by gradual condensation of the chromosomes, formation of the mitotic spindle, and formation of a pair of **centrosomes**, from which microtubules radiate and eventually take up positions at the poles of the cell.
- ***Prometaphase.*** Here, the nuclear membrane dissolves, allowing the chromosomes to disperse within the cell and to attach, by their kinetochores, to microtubules of the mitotic spindle.
- ***Metaphase.*** At this stage, the chromosomes are maximally condensed and line up at the equatorial plane of the cell.
- ***Anaphase.*** The chromosomes separate at the centromere, and the sister chromatids of each chromosome now become independent **daughter chromosomes,** which move to opposite poles of the cell.
- ***Telophase.*** Now, the chromosomes begin to decondense from their highly contracted state, and a nuclear membrane begins to re-form around each of the two daughter nuclei, which resume their interphase appearance. To complete the process of cell division, the cytoplasm cleaves by a process known as **cytokinesis.**

There is an important difference between a cell entering mitosis and one that has just completed the process. A cell in G_2 has a fully replicated genome (i.e., a 4n complement of DNA), and each chromosome consists of a pair of sister chromatids. In contrast, after mitosis, the chromosomes of each daughter cell have only one copy of the genome. This copy will not be duplicated until a daughter cell in its turn reaches the S phase of the next cell cycle (see Fig. 2-8). The entire process of mitosis thus ensures the orderly duplication and distribution of the genome through successive cell divisions.

The Human Karyotype

The condensed chromosomes of a dividing human cell are most readily analyzed at metaphase or prometaphase. At these stages, the chromosomes are visible under the microscope as a so-called **chromosome spread;** each chromosome consists of its sister chromatids, although in most chromosome preparations, the two chromatids are held together so tightly that they are rarely visible as separate entities.

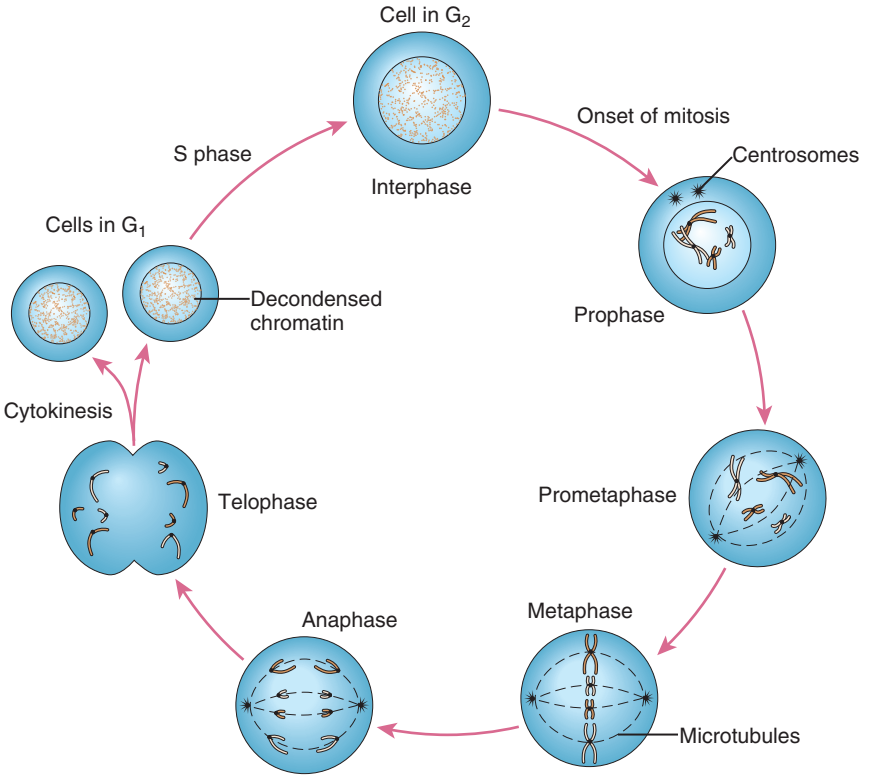


Figure 2-9 **Mitosis.** Only two chromosome pairs are shown. For details, see text.